Which lymphoid cancers should learn from each other?

Request for review of a list fixed before testing

Draft for the study haematologists. Fill in this PDF directly (Adobe Reader or Preview), save it and send it to Mikkel Werling.

Your name:

Background. We have built one prediction model for patients starting first-line treatment for any of 24 lymphoid cancers (37,531 Danish patients). For each patient it predicts the risk of 69 outcomes over the following two years, from relapse and second-line treatment to infection, cytopenia, organ toxicity and hospital admission. Because the model learns from all diseases at once, what it learns in one disease can be used for another.

What we will test. After the model is trained, we can switch off what it learned from one disease and see how much worse its predictions become for another. We want to check whether the model relies on other diseases *where clinical knowledge says it should*. To make this a fair test, we must write down the expected pairs **before** we look at the results on the test patients. Once the test data are opened, the list can no longer be changed. Your clinical judgement now is therefore what makes the test meaningful.

What we ask of you.

1. For each pair below, tick *agree*, *disagree* or *unsure*: would you expect what is learned in the first disease to help predict the stated outcomes in the second? Add a comment where useful. We deliberately give no reasons, so as not to steer your view.
2. Suggest pairs we have missed. Any disease can be paired with any other, for any group of outcomes; the diseases and outcomes are listed at the end. Pairs that would not be obvious from the classification alone, such as diseases that share a treatment or a typical complication, are especially useful.
3. Check the *negative controls* (the last three rows): pairs we expect *not* to inform each other. They show whether the method finds a signal where none should exist.

	Pair	For predicting	Your view	Comment
Pairs expected to inform each other				
1	Learn from **Chronic lymphocytic leukaemia (CLL)** to predict for **Small lymphocytic lymphoma (SLL)**	all outcomes	agree disagree unsure	
2	Learn from **Multiple myeloma (MM)** to predict for **Plasma-cell leukaemia (PCL)**	disease course and later malignancy	agree disagree unsure	
3	Learn from **Extranodal marginal-zone lymphoma (EMZL)** to predict for **Splenic marginal-zone lymphoma (SMZL)**	disease course and later malignancy	agree disagree unsure	
4	Learn from **Diffuse large B-cell lymphoma (DLBCL)** to predict for **Transformed follicular lymphoma (tFL)**	disease course and later malignancy	agree disagree unsure	

	Pair	For predicting	Your view	Comment
5	Learn from **Diffuse large B-cell lymphoma (DLBCL)** to predict for **Richter transformation (RT)**	disease course and later malignancy	agree disagree unsure	
6	Learn from **Chronic lymphocytic leukaemia (CLL)** to predict for **Richter transformation (RT)**	infection and antimicrobial use	agree disagree unsure	
7	Learn from **Multiple myeloma (MM)** to predict for **Chronic lymphocytic leukaemia (CLL)**	infection and antimicrobial use	agree disagree unsure	
8	Learn from **Chronic lymphocytic leukaemia (CLL)** to predict for **Multiple myeloma (MM)**	infection and antimicrobial use	agree disagree unsure	
9	Learn from **Multiple myeloma (MM)** to predict for **Amyloidosis**	renal injury and support	agree disagree unsure	
10	Learn from **Burkitt lymphoma (BL)** to predict for **Lymphoblastic lymphoma (LBL)**	neutrophil, white-cell and platelet findings	agree disagree unsure	
11	Learn from **Burkitt lymphoma (BL)** to predict for **Lymphoblastic lymphoma (LBL)**	metabolic and cell-turnover markers	agree disagree unsure	
12	Learn from **Peripheral T-cell lymphoma, NOS (PTCL)** to predict for **T-cell lymphoma, unspecified (TCL)**	disease course and later malignancy	agree disagree unsure	
13	Learn from **Chronic lymphocytic leukaemia (CLL)** to predict for **Lymphoplasmacytic lymphoma (LPL)**	cardiovascular and thrombotic events	agree disagree unsure	

	Pair	For predicting	Your view	Comment
Negative controls: pairs expected *not* to inform each other				
C1	Learn from **Amyloidosis** to predict for **Hodgkin lymphoma (HL)**	all outcomes	agree disagree unsure	
C2	Learn from **Anaplastic large-cell lymphoma (ALCL)** to predict for **Solitary plasmacytoma**	all outcomes	agree disagree unsure	
C3	Learn from **Transformed follicular lymphoma (tFL)** to predict for **Hairy-cell leukaemia (HCL)**	disease course and later malignancy	agree disagree unsure	

Pairs we have missed (learn from which disease, to predict which outcomes in which disease, and why):

Other comments:

The 24 diseases

Large B-cell lymphomas: Diffuse large B-cell lymphoma (DLBCL); Richter transformation (RT); Transformed follicular lymphoma (tFL)

Burkitt and lymphoblastic lymphomas: Burkitt lymphoma (BL); Lymphoblastic lymphoma (LBL)

Mantle-cell lymphoma: Mantle-cell lymphoma (MCL)

Indolent B-cell lymphomas: Follicular lymphoma (FL); Lymphoplasmacytic lymphoma (LPL); Extranodal marginal-zone lymphoma (EMZL); Splenic marginal-zone lymphoma (SMZL); Nodal marginal-zone lymphoma (NMZL); B-cell lymphoma, unspecified (BCL)

CLL and SLL: Chronic lymphocytic leukaemia (CLL); Small lymphocytic lymphoma (SLL)

Hairy-cell leukaemia: Hairy-cell leukaemia (HCL)

Hodgkin lymphoma: Hodgkin lymphoma (HL)

T-cell lymphomas: Peripheral T-cell lymphoma, NOS (PTCL); Angioimmunoblastic T-cell lymphoma (AITL); Anaplastic large-cell lymphoma (ALCL); T-cell lymphoma, unspecified (TCL)

Plasma-cell neoplasms: Multiple myeloma (MM); Plasma-cell leukaemia (PCL); Solitary plasmacytoma

Amyloidosis: AL amyloidosis

The 69 outcomes, by clinical domain

Each outcome is predicted at 30, 60 and 90 days and at 6, 12 and 24 months after the start of first-line treatment. Laboratory grades are based on the CTCAE laboratory thresholds; pairs can name a whole domain or "all outcomes".

Disease course and later malignancy: overall survival; second-line treatment; treatment failure; treatment failure or transformation; new haematological malignancy; new malignancy (any)

Infection and antimicrobial use: sepsis; serious infection (including invasive fungal infection and pneumocystosis); antibiotic treatment; intravenous antimicrobial use; prescription antimicrobial use; infection diagnosis or antimicrobial use

Erythroid findings and transfusion: anaemia grade 2 and 3; red-cell transfusion; any transfusion; haptoglobin decreased

Neutrophil, white-cell and platelet findings: neutropenia grade 2 and 3; leukopenia grade 2 and 3; thrombocytopenia grade 2 and 3; platelet transfusion; severe cytopenia

Lymphocyte and eosinophil counts: lymphocytosis grade 2 and 3; eosinophilia

Renal injury and support: acute kidney injury; creatinine increased grade 2 and 3; creatinine rise above 1.5 times the pre-treatment value; dialysis

Liver injury and coagulation: ALT increased grade 2 and 3; alkaline phosphatase elevation, two levels; bilirubin increased grade 2 and 3; hepatotoxicity; INR increased grade 2 and 3

Sodium and potassium: hyponatraemia grade 2; sodium below 125 mmol/L; potassium below 3.0 mmol/L; hypernatraemia grade 2 and 3; hyperkalaemia grade 2 and 3

Calcium and magnesium: hypercalcaemia grade 2 and 3; hypocalcaemia grade 2 and 3; hypomagnesaemia grade 2

Metabolic and cell-turnover markers: hypertriglyceridaemia grade 2 and 3; hypoalbuminaemia grade 2 and 3; hyperphosphataemia; hyperuricaemia; LDH increased

Cardiovascular and thrombotic events: atrial fibrillation; heart failure; stroke or transient ischaemic attack; venous thromboembolism

Acute care and other events: acute hospitalisation; emergency department stay; respiratory support; bone fracture